Phyloseq Visualizations in Galaxy

BIOL 4810

2026-04-16

Goal: Use **phyloseq in Galaxy** to visualize your microbial community data with **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

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0) Your dataset

0A. Sample set / project name:

WCSP Sample

0B. Grouping variable used for comparisons:

Sample Source and Infection Status

0C. Other metadata variable(s) available for plotting or faceting:

Sex, parasite load, polychromasia, and HL

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phyloseq workflow focus

For this module, you will generate and interpret:

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Alpha diversity → Beta diversity (NMDS using Bray-Curtis) → Taxonomic
bar plot
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Reminder: These visualizations help you describe diversity within samples, differences among samples, and which taxa are driving those patterns.

Part 1 — Alpha diversity

1A) Alpha diversity

In your own words, what is alpha diversity?

Alpha diversity is the total diversity within a single dataset, demonstrating the various taxa that are present and how they are distributed in a dataset.

1B) Observed richness

What does Observed measure?

Observed richness measures the presence, or total number, of ASVs that are detected in a single sample.

1C) Shannon diversity

What does Shannon measure?

The Shannon diversity measures the richness and evenness of different taxa.

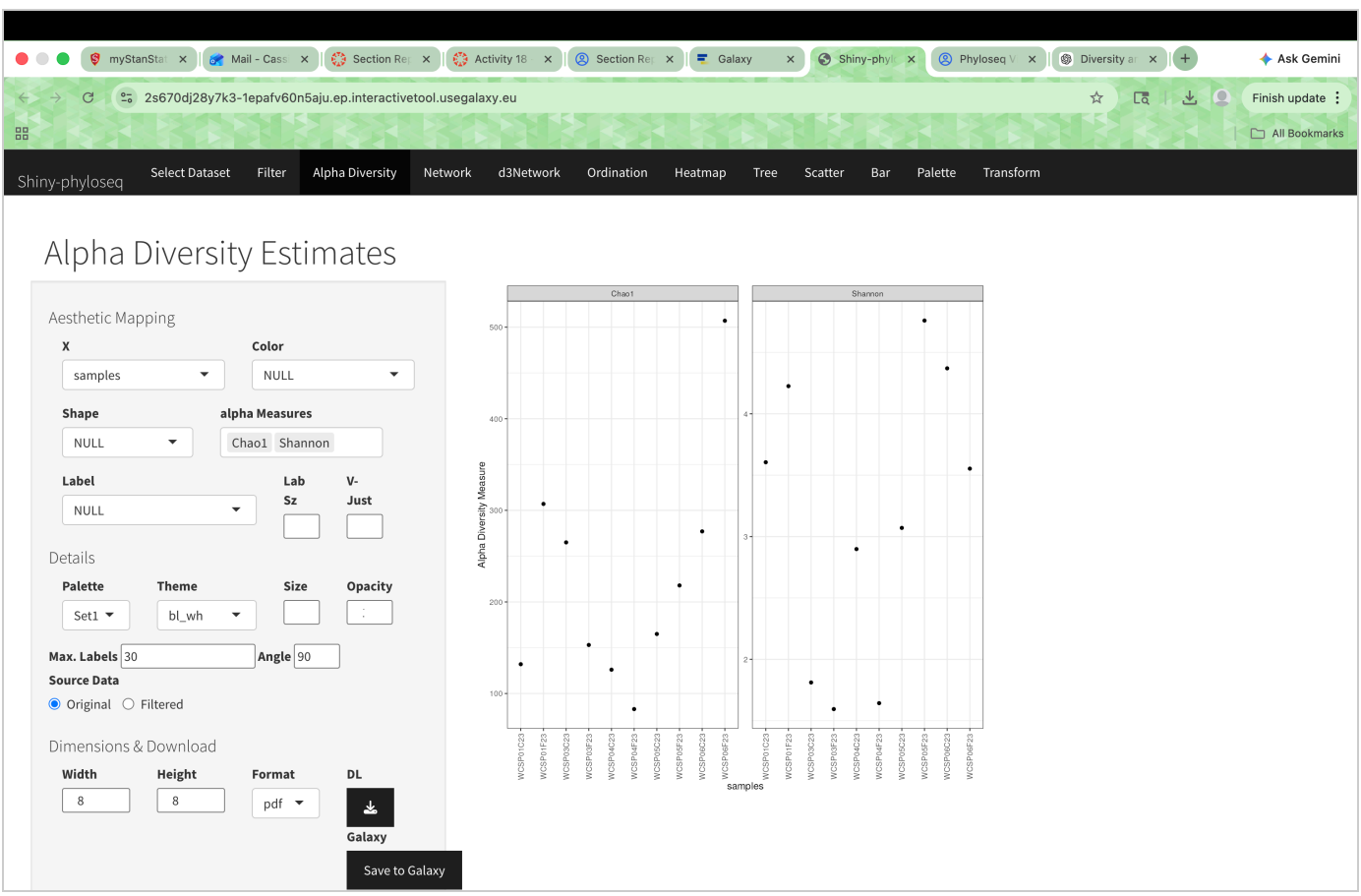
1D) Simpson diversity

What does Simpson measure?

The Simpson diversity measures the likelihood that two random individuals are in different taxa, thus measuring the dominance of species.

Upload screenshot: alpha diversity plot

Choose File Screenshot …10.30.45 AM



Grouping variable shown on the alpha diversity plot:

Samples

Which sample or group appears to have the greatest alpha diversity?

WCSP06F23

Did Observed, Shannon, and Simpson show the same pattern?

These three metrics do not exactly demonstrate the same pattern, but are relatively similar, because some of the samples with high observed values do not demonstrate the highest Shannon values. This suggests that some

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Part 2 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:
Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

Beta diversity is the variety in the microbes between samples, demonstrating how conditions are both similar and different across various conditions.

2A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

Bray-Curtis measures the samples based on the taxa abundance, ranking how different microbial communities are from one another, ranking them on a scale of 0 (not identical) to 1 (identical).

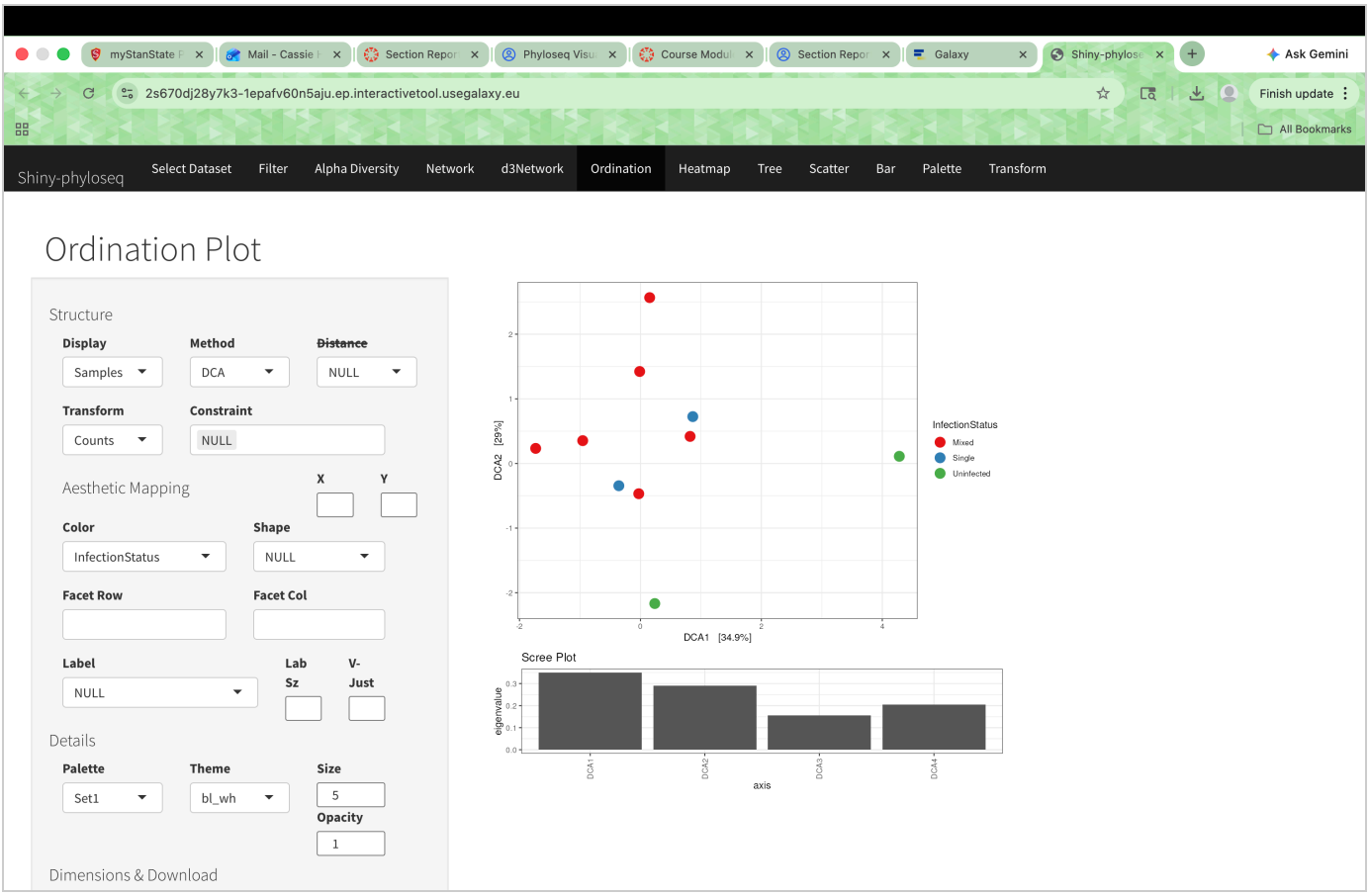
2B) NMDS

What does NMDS show?

An NMDS plot demonstrates the similarities among samples in a 2D plot. Samples that are closer to one another are more similar to one another.

Upload screenshot: NMDS Bray-Curtis plot

Choose File Screenshot …10.25.00 AM



Color variable used on the NMDS plot:

InfectionStatus

Shape variable used on the NMDS plot (if any):

N/A

Which samples or groups cluster together?

The single samples clustered together, the mixed samples clustered together, and the uninfected samples clustered together.

Are any groups clearly separated from one another? What might that suggest?

The uninfected samples are clustered together, which might suggest that their microbial communities may be closer to one another than other groups.

If points overlap strongly, what does that mean?

If points strongly overlap, then this indicates that the samples have close microbial community compositions. Therefore, the factor that is being compared (e.g. InfectionStatus) may not have a strong effect on the structure

Part 3 — Taxonomic bar plot

What a taxonomic bar plot shows:
A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

Phylum

How did you facet the plot?

Facet by variable (SampleSource or InfectionStatus)

Why was this taxonomic level useful?

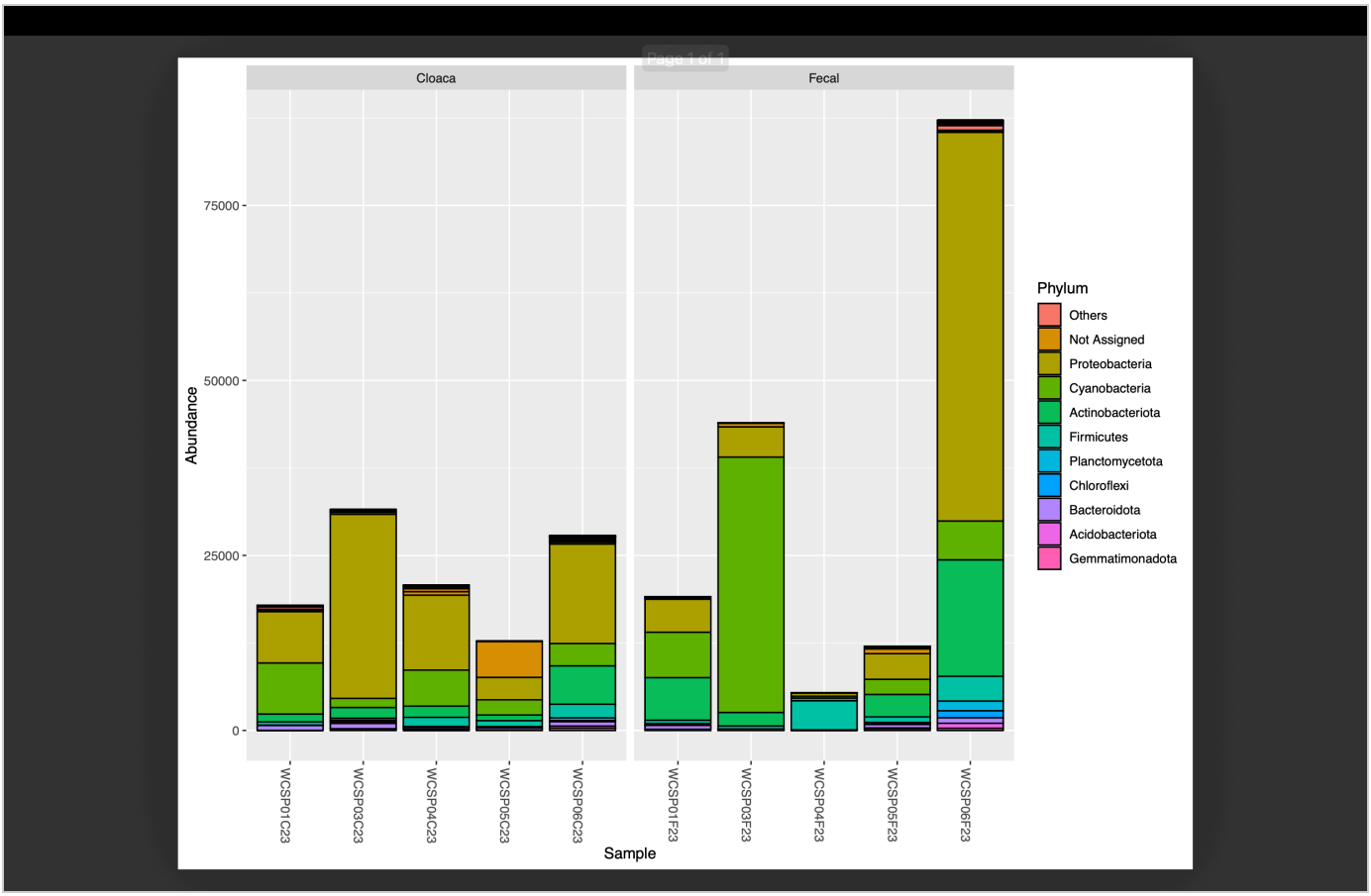
This taxonomic level is useful because it indicates which major microbes are present.

Why was your faceting choice useful?

My faceting choice was useful because it makes the differences between groups easier to detect, enabling us to better compare the groups.

Upload screenshot: taxonomic bar plot

Choose File Screenshot …t 9.55.32 AM



Which taxa appear to be most abundant?

The taxa that appears to be the most abundant is proteobacteria

Did taxonomic composition differ among groups?

The taxonomic composition did differ among groups. Even though proteobacteria was the most prominent across the different groups, cyanobacteria had a relative presence in every group, as well as actinobacter. The

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Part 4 — Overall interpretation

In 3–5 sentences, summarize what your alpha diversity, beta diversity, and bar plot together suggest about this dataset.

These results demonstrate that even though the overall diversity between the groups (which is indicated by the alpha diversity) may differ between the groups, the community composition between the groups (which is indicated by the beta diversity) demonstrates clearer separation. Therefore, this suggests that there are distinct assemblages of microbes. The clustering that is shown in the beta diversity plot show that the samples in the same group are more similar to each other than those in the other groups. The taxonomic bar plot supports this

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Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset
Project/sample set: WCSP Sample
Grouping variable: Sample Source and Infection Status
Other metadata variables: Sex, parasite load, polychromasia, and HL

Alpha diversity
Definition: Alpha diversity is the total diversity within a single dataset, demonstrating the various taxa that are present and how they are distributed in a dataset.
Observed: Observed richness measures the presence, or total number, of ASVs that are detected in a single sample.
Shannon: The Shannon diversity measures the richness and evenness of different taxa.
Simpson: The Simpson diversity measures the likelihood that two random individuals are in different taxa, thus measuring the dominance of species.
Grouping variable: Samples
Highest alpha diversity: WCSP06F23
Did the metrics show the same pattern? These three metrics do not exactly demonstrate the same pattern, but are relatively similar, because some of the samples with high observed values do not demonstrate the highest Shannon values. This suggests that some taxa are dominated by a few specie.

Beta diversity (NMDS Bray-Curtis)
Definition: Beta diversity is the variety in the microbes between samples, demonstrating how conditions are both similar and different across various conditions.
Bray-Curtis: Bray-Curtis measures the samples based on the taxa abundance, ranking how different microbial communities are from one another, ranking them on a scale of 0 (not identical) to 1 (identical).
NMDS: An NMDS plot demonstrates the similarities among samples in a 2D plot. Samples that are closer to one another are more similar to one another.
Color variable: InfectionStatus
Shape variable: N/A
Which groups cluster together? The single samples clustered together, the mixed samples clustered together, and the uninfected samples clustered together.
Are any groups separated? The uninfected samples are clustered together, which might suggest that their microbial communities may be closer to one another than other groups.
If points overlap strongly: If points strongly overlap, then this indicates that the samples have close microbial community compositions. Therefore, the factor that is being compared (e.g. InfectionStatus) may not have a strong effect on the structure of the community.

Taxonomic bar plot
Taxonomic level: Phylum
Faceted by: Facet by variable (SampleSource or InfectionStatus)
Why this taxonomic level? This taxonomic level is useful because it indicates which major microbes are present.
Why this faceting choice? My faceting choice was useful because it makes the differences between groups easier to detect, enabling us to better compare the groups.
Most abundant taxa: The taxa that appears to be the most abundant is proteobacteria
Did composition differ among groups? The taxonomic composition did differ among groups. Even though

proteobacteria was the most prominent across the different groups, cyanobacteria had a relative presence in every group, as well as actinobacter. The taxa that are pink and purple are less prevalent compared to the other groups.

Overall interpretation

These results demonstrate that even though the overall diversity between the groups (which is indicated by the alpha diversity) may differ between the groups, the community composition between the groups (which is indicated by the beta diversity) demonstrates clearer separation. Therefore, this suggests that there are distinct assemblages of microbes. The clustering that is shown in the beta diversity plot show that the samples in the same group are more similar to each other than those in the other groups. The taxonomic bar plot supports this evidence by exhibiting the shifts in the abundance of key taxa across the groups.